

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 15: Iris Flower Classification using Naive Bayes Classifier

Aim

To implement the Naive Bayes classifier in Python for classifying Iris flowers into different species.

Procedure

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the features and target classes.
4. Split the dataset into training and testing sets.
5. Create a Gaussian Naive Bayes classifier.
6. Train the classifier using the training data.
7. Calculate the accuracy and confusion matrix.
8. Predict the class of a new Iris flower.

Algorithm

1. Start.
2. Load the Iris dataset.
3. Divide the dataset into input features and target classes.
4. Split the data into training and testing sets.
5. Initialize the Gaussian Naive Bayes classifier.
6. Train the classifier using the training data.
7. Predict the class labels of the test data.
8. Calculate the classification accuracy.
9. Predict the class of a new flower.
10. Stop.

Dataset 1

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
5.8	2.7	4.1	1.0	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.5	3.0	5.2	2.0	Virginica

Dataset 2

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.0	3.4	1.5	0.2	Setosa
5.2	3.5	1.5	0.2	Setosa
5.7	2.8	4.1	1.3	Versicolor
6.1	2.8	4.7	1.2	Versicolor
6.7	3.1	5.6	2.4	Virginica

Dataset 3

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
4.8	3.1	1.6	0.2	Setosa
5.4	3.7	1.5	0.2	Setosa
5.9	3.0	4.2	1.5	Versicolor
6.2	2.9	4.3	1.3	Versicolor
6.4	2.8	5.6	2.1	Virginica

Dataset 4

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.1	3.8	1.5	0.3	Setosa
4.7	3.2	1.3	0.2	Setosa
6.2	2.2	4.5	1.5	Versicolor
5.6	2.9	3.6	1.3	Versicolor
6.9	3.1	5.4	2.1	Virginica

Dataset 5

Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Species
5.0	3.6	1.4	0.2	Setosa
5.3	3.7	1.5	0.2	Setosa
5.5	2.4	3.8	1.1	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.5	3.0	5.2	2.0	Virginica

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:")
print(y_test)

print("Predicted Labels:")
print(y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("New Flower Classification:", data.target_names[prediction[0]])
```

Output

```
... Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
New Flower Classification: setosa
```

Result

The Iris Flower Classification was successfully implemented using the Gaussian Naive Bayes classifier. The model classified Iris flowers into Setosa, Versicolor, and Virginica with high accuracy.